

SUBJECT : PROBLEM SOLVING AND TESTING USING JAVA

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COURSE CODE : 10211CS227

VTU NUMBER : VTU27544

Task 2 (Difficult): Real-Time Stream Analytics Engine

Problem Description

A cloud analytics platform receives millions of sensor readings. Each reading contains a sensor ID and temperature value. Using stream processing concepts, perform the following operations:

1. Filter temperatures greater than 50.
2. Group readings by sensor ID.
3. Compute average temperature per sensor.
4. Sort sensors based on average temperature in descending order.

Input Format

- First line contains integer N.
- Next N lines contain SensorID and Temperature.

Output Format

Display SensorID and average temperature sorted in descending order.

Constraints

- $1 \leq N \leq 10^5$
- $0 \leq \text{Temperature} \leq 100$

Sample Input

```
6
S1 60
S2 40
S1 80
S3 70
```

S2 90

S3 30

Sample Output

S1 70.0

S2 90.0

S3 70.0

PROGRAM :

```
import java.util.*;
```

```
import java.util.stream.*;
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int n = sc.nextInt();
```

```
        Map<String, List<Double>> data = new HashMap<>();
```

```
        for (int i = 0; i < n; i++) {
```

```
            String sensor = sc.next();
```

```
            double temp = sc.nextDouble();
```

```
            // Filter temperature > 50
```

```
            if (temp > 50) {
```

```

        data.computeIfAbsent(sensor, k -> new
ArrayList<>()).add(temp);
    }
}

// Calculate average and sort in descending order
data.entrySet().stream()
    .map(e -> new AbstractMap.SimpleEntry<>(
        e.getKey(),
        e.getValue().stream()
            .mapToDouble(Double::doubleValue)
            .average()
            .orElse(0)
    ))
    .sorted((a, b) -> Double.compare(b.getValue(), a.getValue()))
    .forEach(e -> System.out.println(
        e.getKey() + " " + e.getValue()
    ));

sc.close();
}
}

```

OUTPUT :

Problems116 · Output Debug Console Terminal Ports

```
PS c:\Users\surya\Desktop\New folder> & '(:\Program File<,\Java\jdk-26.0.2\bin\j,wa.c>xe' '- enable-pr
c>view' ' XX: □showcodeoet ail">InE xceptionMes<,age'>' ' cp' 'c: \Us\rs\<,urya\AppData alRoaming\c ode\User\work
spacc>Storage\fb83cd0feddlaJdbb nad6ff7a39h44\redh,1t. j,1va\ jdt ws\New folder_ J771Jc17\bin' 'Main'
6
s1 40
" 50
s1 60
s3 20
s2 30
"40
SI 60.0
PS c: \Users\surya\Desktop\New folder> |
```